

When the object has to be recognized, the features are detected along with their positions and alignments and searched in the database.

Invariance

This technique applies to those properties of an object which are not subject to a change with the position of camera being changed. The condition applies mostly to images of planar objects.

There are a few other recognition techniques such as Geometric Hashing, Pose clustering, and Speed Up Robust Features (SURF). Unfortunately their understanding requires some complex mathematical theories which we cannot go in for the purposes of this FastTrack.

Challenges of Object Recognition

The biggest challenge in Object Recognition happens to be the wide array of possibilities available during the pattern search. In all the techniques we mentioned, a database with all the known object models is needed. However a database of object models of one technique do not serve well for others. For example, the database for searching through features in an image will be different from the one which uses grayscale images for comparison.

Databases would again have to store the images of different objects from different angles in various formats for supporting more than one type of algorithm for pattern matching. While it increases the time taken for object recognition, it improves the accuracy and brings down the probability of error because none of the techniques are deemed perfect.

Further difficulty is posed by the requirement to identify partly visible objects. Going back to our case of detecting a house - from some viewpoints, it is possible to achieve only limited visibility of the house - e.g. if you look at a house from the window of another house, you may get only limited visibility of the object. When the visibility of an object is obstructed, identification becomes more difficult. For this, the comparison algorithm sitting between the camera and object database has to be more clever.

In many cases when more than one objects are present in front of the camera, one object would obstruct the view of another. Under such circumstances, it is needed to detect both objects and figure out which object is obstructing the view of another. The simple method for such comparisons is to see which object provides a non-obstructing (complete) view and which one does not. Adding more objects in the mix may eventually bring out an erratic behavior from the algorithm.